

# Deepesh Sonar

I build systems, then try to break my own claims about them. · Mumbai, India

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## EDUCATION

B.E. Computer Engineering, Thakur College of Engineering and Technology (TCET), Mumbai — expected 2028 · CGPA 9.53 / 10

## FLAGSHIP PROJECTS

### ICE — Infinite Context Engine (Jun 2026 — Present) — research · manuscript complete, follow-up eval ongoing

A local-first memory layer for conversational AI: persistent, structured memory for any OpenAI-compatible model, running entirely on your own hardware.

- Structured episodic, document, procedural and temporal-graph stores behind an OpenAI-compatible proxy.
- Pre-flight classifier: multi-head PyTorch head over frozen Qwen3-Embedding-0.6B — 27 labels across topic, intent and context-reliance; calibrated routing decides when retrieval fires at all.
- Hybrid retrieval (lexical · vector · graph · procedural · document · timeline) with rank fusion, dedup and bounded token budgets.

### Presentation Forge (Aug — Sep 2026) — active · self-hosted

Self-hosted research-to-artifact pipeline: topic or sources in, researched, themed, editable PPTX/DOCX and report out. Docker-first, BYOK or Ollama.

- 74 slide types · 34 themes with deterministic rendering and schema validation.
- Human-approval gates, critique pass and reproducible reports with automated checks.
- Local-model support (Ollama) or BYOK providers; no inference bill required.

### Timetable Generator (Mar 2026 — Present) — active development

Constraint-driven scheduling service for colleges: rooms, faculty, groups and subjects in; ranked, publishable timetables out.

- Greedy + OR-Tools CP-SAT solvers under explicit hard constraints and weighted soft-constraint scoring.
- Authenticated REST APIs with audit/publishing workflows, shared-resource reservations and cross-timetable safety.
- Data-driven constraint registry; no post-generation cleanup hacks.

### Prompt Routing Classifier (Mar 2026) — complete · basis for ICE router

Multi-label topic + intent classifier for routing prompts to specialised models: sentence embeddings + one-vs-rest logistic regression, CPU-only.

- 5,100-prompt labelled corpus with documented dataset, training path and evaluation split.
- Held-out F1 0.68 (topic) / 0.56 (intent); CPU-only inference.
- Routing design reused as the basis for ICE's pre-flight classifier.

### NEXUS Fraud Detection (team · my PR merged) (Sep 2026) — team · expansion PR merged upstream

Fraud-detection platform (transactions + messages + URLs). I took the deterministic-rules foundation and added a full transaction fraud head, a stronger message head, a debiased URL head, website flows and an admin-swappable AI provider — while removing the WhatsApp/n8n dependency.

- PR +3,455/" 1,228 across 59 files, merged upstream; trains and serves from the repo alone.
- Transaction head (XGBoost + Optuna + isotonic calibration, grouped User-Card split): test PR-AUC 0.707; fresh-seed check 0.689 — reported in the PR with eval ledger.
- Message head (char+word TF-IDF + direction features + augmentation): 11/18 !' 18/18 on unseen probes; URL head debiased (bare-domain + www-bias fixes + ~30k synthetic benign-with-path rows): 12/20 !' 19/20.

## RESEARCH

### LSREP: A Longitudinal State-Replay Protocol for Evaluating Conversational Memory, with ICE v2 as an Audited Local-First Architecture

arXiv cs.AI · arXiv:2609.16730 — LSREP treats memory as evolving over time — ordered replay, repeated probes, evolving references, lifecycle schedules, mechanism-fidelity auditing. Grew out of building ICE; the larger question is how you know a system behaves the way its architecture says it does. <https://arxiv.org/abs/2609.16730>

### ICE v2 manuscript (37 pp.)

manuscript — complete, in revision — Frozen v2 snapshot: controlled evaluation + matched LongMemEval diagnostic. Further v3 evaluation is follow-up engineering, not a prerequisite for describing the paper. <https://github.com/Deepnar/ice>

## MERGED UPSTREAM

- OpenAgentHQ/modeldock — map Jan runtime capabilities (<https://github.com/OpenAgentHQ/modeldock/pull/221>)
- OpenAgentHQ/modeldock — discover existing GPT4All model directory (<https://github.com/OpenAgentHQ/modeldock/pull/222>)
- vllm-project/semantic-router — default Envoy logging to info (<https://github.com/vllm-project/semantic-router/pull/3288>)

- mne-tools/mne-python — avoid pytest 10 warnings in sklearn estimator checks (<https://github.com/mne-tools/mne-python/pull/14283>)
- Newer1107/CivicResolve — AI Observation Engine (street-imagery verification, worker, tests, docs) (<https://github.com/Newer1107/CivicResolve/pull/1>)

## **TEAM WORK (MY LANE)**

SIH 2025 ×2 rounds · DIPEX 2026 state final · NEXUS (merged: fraud/message/URL heads, txn flows, AI-provider console) · RapidRail (core: booking, wallet, QR tickets) · IIS-mini (Person-1 classical baselines).

## **STACK**

Python · PyTorch · FastAPI · PostgreSQL + pgvector · Docker · Linux · Java · Rust (learning) · OR-Tools · Ollama